

Technology #12: Osmotic Pressure Regulating Drainage Valve

Technology Domain: Osmotic Membrane Transport + Membrane Mechanics

Mechanism: CSF shunt valve regulated by osmotic pressure differential across semipermeable membrane. Osmotic agent concentration on one side of membrane modulates effective drainage pressure based on CSF protein concentration or implanted osmotic reservoir.

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: Membrane fouling rate in CSF reduces L_p by >50% within 30 days (insufficient for chronic implant)

Next Experiment: Membrane specimens (various materials + geometries)

Buyer Receives: Osmotic regulation concept + mechanism description; Van 't Hoff + Kedem-Katchalsky governing equations; Bench test protocols (permeation, fouling, aging, drainage); External precedent catalog (osmotic pumps: DURECT, Alzet); Critical UNKNOWN disclosure (membrane fouling rate, reservoir depletion rate); This engineering dossier with failure analysis + build plan

Buyer Must Build: Production membrane design (material, geometry, L_p , σ); Osmotic reservoir design (agent, volume, lifetime); Manufacturing process for membrane + reservoir + valve integration; Regulatory submission (likely Class III PMA — no predicate for osmotic CSF valve); Clinical validation (drainage performance endpoint); Long-term fouling + depletion data

Transfer Posture: Sponsored Validation

Target Buyer: Neuroshunt manufacturer; membrane technology company

CONFIDENTIAL. See full engineering dossier for details.